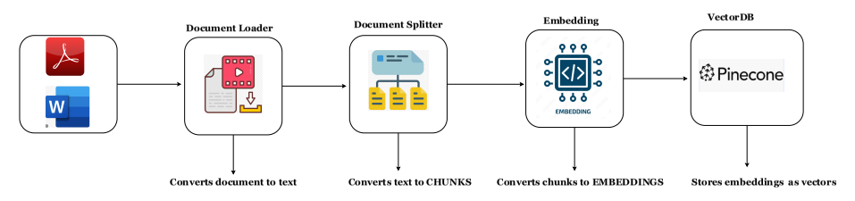
Need - Of - Chunking - Strategies

Ref: https://www.pinecone.io/learn/chunking-strategies/

Process of Converting documents to vectors



Converts document to text

Converts text to CHUNKS

Converts chunks to EMBEDDINGS

Stores embeddings as vectors

Problem with Chunking

Given Data

Employees are eligible for reimbursement of travel expenses including airfare, hotel, and meals.

However, reimbursement is only approved if the total expense is less than $5000 and the trip was pre-approved by the manager.

If the expense exceeds $5000, special approval from the finance director is required before reimbursement can be processed.

Chunk1
Employees are eligible for reimbursement of travel expenses including airfare, hotel, and meals. However, re

Chunk2
imbursement is only approved if the total expense is less than $5000 and the trip was pre-approved by the ma

Chunk3
nager. If the expense exceeds $5000

Chunk4
pecial approval
from the finance director is required before reimbursement can be processed

Question

Is approval required for expenses above $5000?

The relevant information is split across:
Chunk 3 → "expense exceeds $5000"
Chunk 4 → "approval from finance director is required"
But Chunk 3 does NOT contain the complete meaning.
Chunk 4 alone does NOT mention $5000.

Result: Retrieval failure

Chunk 3:
"If the expense exceeds $5000, special approval from the finance director is required before reimbur"

LLM might hallucinate or give incomplete answer like:
Approval may be required.
Instead of correct answer:
Yes, approval from finance director is required.

Solution

Chunking means → Breaking a large document into smaller pieces

Solution → chunking is breaking large text into smaller segments and says the key is to create chunks that are large enough to contain meaningful information,
but small enough for performant retrieval and low-latency applications.

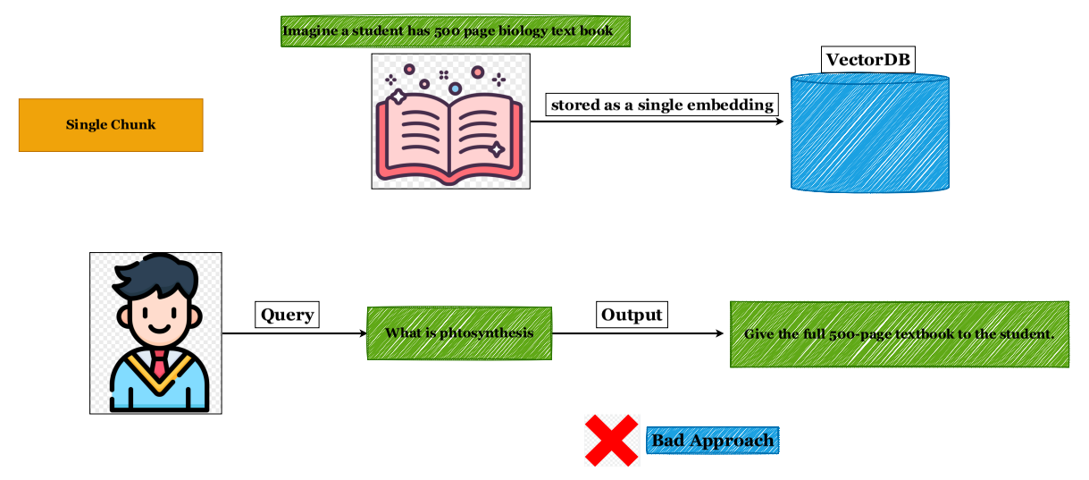
Hence its mandatory that you should know Chunking Strategies

Real Life Examples

1. Text Book Example

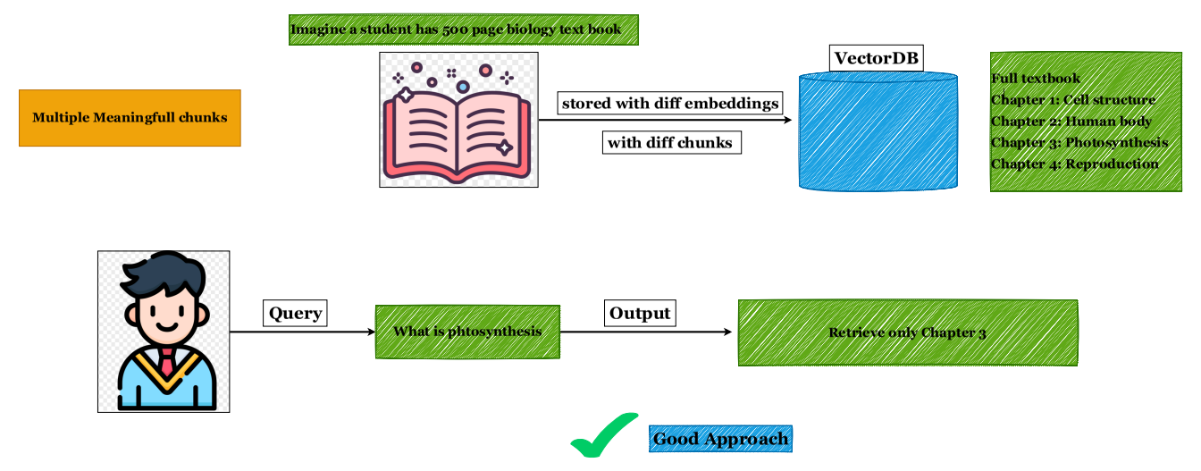
Single Chunk

Imagine a student has 500 page biology text book


stored as a single embedding
VectorDB
Query
What is phtosynthesis
Output
Give the full 500-page textbook to the student.
Bad Approach

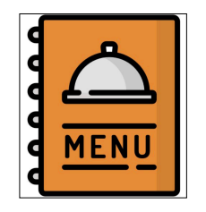
Multiple Meaningfull chunks

Imagine a student has 500 page biology text book



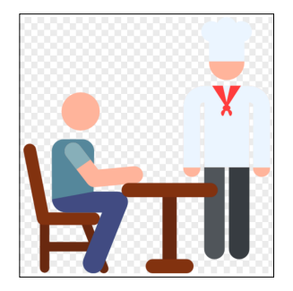
Full textbook
Chapter 1: Cell structure
Chapter 2: Human body
Chapter 3: Photosynthesis
Chapter 4: Reproduction

2. Restaurant Menu



Restaurant Menu

Starters
Soups
Chicken dishes
Vegetarian dishes
Seafood
Desserts
Drinks



Do you have spicy chicken?

With out Chunking ! → Search the entire menu

With chunking ! →
Chunk 1: Starters
Chunk 2: Soups
Chunk 3: Chicken dishes
Chunk 4: Vegetarian dishes
Chunk 5: Desserts

Chunk 3: Chicken Dishes

System will directly retrieves

3. News Recommender Session

Large news article:
- Introduction about Obama
- Healthcare reform
- Economy
- Foreign policy
- Election campaign

User asks:
"What did Obama say about healthcare?"

Without chunking:
The full article is searched as one vector.

With chunking:
The healthcare paragraph becomes its own vector.

Result:
The system retrieves the exact healthcare chunk.

So Finally In short

In real GenAI/RAG applications, we usually deal with large content:

News articles
PDF files
Legal contracts
HR policies
Banking documents
Technical manuals
Customer support documents
Research papers

Now imagine one article has 20 pages, or one policy document has 100 pages.
If we send the entire document directly to the embedding model or LLM, we face multiple problems:

1. The text may exceed the embedding model token limit.
2. Important information may get truncated.
3. The embedding may become too broad and noisy.
4. Search results may become inaccurate.
5. The LLM may receive too much irrelevant context.
6. Response latency and cost may increase.

So chunking Strategies plays an important role in solving this problem